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KPK BOARD TEXTBOOK NOTES - MATHEMATICS CLASS 9

# Unit 15: Pythagoras Theorem

Complete, easy English notes based on *A Textbook of Mathematics for Grade IX*, Khyber Pakhtunkhwa Textbook Board (KPTB), Peshawar - the theorem and its proof idea, finding any side of a right-angled triangle, the converse, Pythagorean triples, and everyday applications, with worked examples and exam practice.

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> **How to use these notes:** This is a short unit with only one main formula, so learn it perfectly. In every question, first find the right angle and name the hypotenuse. Only then decide whether to add the squares or subtract them. That one habit prevents almost every mistake in this unit.

## What this unit covers

1. The right-angled triangle - hypotenuse, perpendicular and base
2. Pythagoras theorem and why it is true
3. Finding the hypotenuse
4. Finding a shorter side
5. The converse of the theorem and Pythagorean triples
6. Applications - ladders, diagonals, distances and heights
7. Worked mixed examples in board style
8. Formula summary, common mistakes, MCQs and practice questions with answers

Source book: KPTB Mathematics for Grade IX, Unit 15 (book pages 285-290). The notes follow the book's order and syllabus. The scan used for verification carries no printed copyright page, so no printing year is claimed here - the 17-unit structure is the current circulating KPTB edition.

# 1. The Right-Angled Triangle

A **right-angled triangle** is a triangle with one angle equal to 90 degrees. The small square drawn in the corner marks that right angle. The three sides have special names, and using the right name is the first step in every question.

- The **hypotenuse** is the side opposite the right angle. It is always the **longest** side of the triangle.
- The other two sides are the **perpendicular** (the vertical side) and the **base** (the horizontal side). Together they are called the two shorter sides, or the legs.
- The hypotenuse never touches the right angle. If you stand at the right-angle corner, the side you face across the triangle is the hypotenuse.

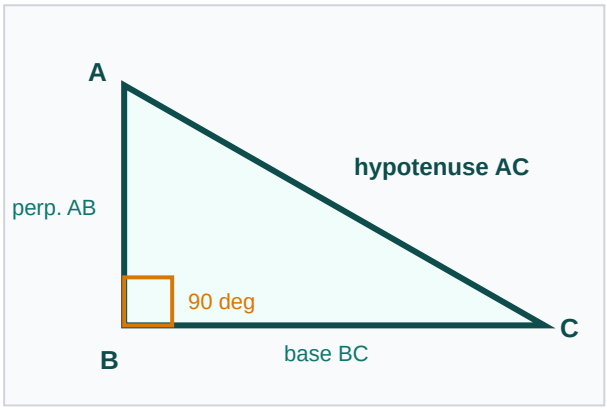



Figure 1. Triangle ABC is right-angled at B. AC, opposite the right angle, is the hypotenuse.

## Letters used in these notes

Throughout this unit, **c** stands for the hypotenuse and **a** and **b** stand for the two shorter sides. Some books use different letters for different triangles, but the rule never changes: the letter for the hypotenuse is always the side opposite the 90-degree angle, wherever the triangle is drawn.

> **Exam trap:** The hypotenuse is not always the sloping side on the right, and it is not always labelled c in the question. Turn the page in your mind if needed: find the 90-degree corner first, then look across from it. If the triangle has no right angle, Pythagoras theorem does not apply at all.

# 2. Pythagoras Theorem - Statement and Proof Idea

The theorem is named after the Greek mathematician **Pythagoras**. It connects the three sides of every right-angled triangle, and only of a right-angled triangle.

**Pythagoras theorem:** In a right-angled triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides.

$c^2 = a^2 + b^2$ where c is the hypotenuse and a, b are the two shorter sides.

Read the formula in words as well as in symbols: *hypotenuse squared equals perpendicular squared plus base squared.* A quick check with the famous 3-4-5 triangle shows it working: $3^2 + 4^2 = 9 + 16 = 25 = 5^2$. The longest side, 5, is the hypotenuse, and its square balances the other two squares exactly.

## Why it is true - the area idea

Draw squares on all three sides of a right-angled triangle. The theorem says that the **area of the square on the hypotenuse** is exactly equal to the **areas of the other two squares added together**. One classic proof cuts the two smaller squares into pieces and rearranges those pieces, without any gap or overlap, to fill the large square on the hypotenuse exactly. Because the pieces fill it perfectly, the areas must be equal - and the side lengths must satisfy $c^2 = a^2 + b^2$.

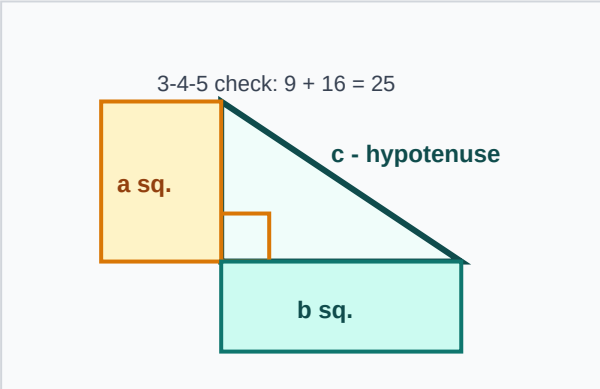



Figure 2. Squares on the three sides: the area on the hypotenuse equals the other two areas added.

**What the theorem gives you:** If any two sides of a right-angled triangle are known, the third side can always be calculated. Sections 3 and 4 are the two cases: finding the hypotenuse, and finding a shorter side.

# 3. Finding the Hypotenuse

When the two shorter sides are known and the hypotenuse is missing, **add** the squares. The hypotenuse is the longest side, so its square must be larger than each single square - addition is the operation that makes it larger.

**Hypotenuse:** $c = \text{sqrt}( a^2 + b^2 )$ - square both shorter sides, add, then take the square root.

**Example 1.** The two shorter sides of a right-angled triangle are 6 cm and 8 cm. Find the hypotenuse.

$c^2 = 6^2 + 8^2 = 36 + 64 = 100$. So c = sqrt(100) = **10 cm**. Check: 10 is longer than both 6 and 8, as a hypotenuse must be.

**Example 2.** A right-angled triangle has perpendicular 5 m and base 12 m. Find the hypotenuse.

$c^2 = 5^2 + 12^2 = 25 + 144 = 169$. So c = sqrt(169) = **13 m**. The numbers 5, 12 and 13 are worth remembering - they appear again in Section 5.

**Example 3 - hypotenuse not a whole number.** Find the hypotenuse when the shorter sides are 2 cm and 3 cm.

$c^2 = 2^2 + 3^2 = 4 + 9 = 13$. So c = **sqrt(13) cm**, about 3.61 cm. Not every answer is a whole number: leave sqrt(13) as the exact answer unless the question asks for a decimal.

## Safe working routine

1. Write $c^2 = a^2 + b^2$ first - the examiner awards a mark for the correct formula.
2. Substitute the two given sides and work out each square separately.
3. Add, and only then take the square root. Never take the root of each square and add those - sqrt(36 + 64) is 10, not 6 + 8 = 14.
4. Sanity-check: the answer must be longer than each given side, but shorter than the two sides added together.

# 4. Finding a Shorter Side

When the hypotenuse and one shorter side are known, rearrange the theorem. The missing shorter side is found by **subtracting** the squares - never by adding, because a shorter side must come out smaller than the hypotenuse.

**Shorter side: $a = \sqrt{c^2 - b^2}$**

From $c^2 = a^2 + b^2$ we get $a^2 = c^2 - b^2$. Square the hypotenuse, subtract the square of the known shorter side, then take the square root.

**Example 4.** The hypotenuse of a right-angled triangle is 10 cm and one shorter side is 6 cm. Find the other shorter side.

$a^2 = 10^2 - 6^2 = 100 - 36 = 64$. So $a = \sqrt{64}$ = **8 cm**. Notice this is the 6-8-10 triangle of Example 1, worked backwards.

**Example 5.** A right-angled triangle has hypotenuse 17 cm and base 15 cm. Find its perpendicular.

Perpendicular$^2 = 17^2 - 15^2 = 289 - 225 = 64$. So the perpendicular = $\sqrt{64}$ = **8 cm**. The sides 8, 15 and 17 form another family worth remembering.

**Example 6.** The hypotenuse is 25 m and one side is 7 m. Find the remaining side.

Other side$^2 = 25^2 - 7^2 = 625 - 49 = 576$. So the other side = $\sqrt{576}$ = **24 m**. Check by adding: $7^2 + 24^2 = 49 + 576 = 625 = 25^2$. Correct.

**The add-or-subtract decision:** Missing the hypotenuse? Add. Missing a shorter side? Subtract. Decide this before you touch the numbers, and write the rearranged formula on its own line. Adding when you should subtract is the single commonest error in this unit, and it always shows itself: the answer comes out longer than the hypotenuse, which is impossible.

**Units and sense:** Keep the same unit (cm, m) through the whole question, and finish with that unit in the answer. A length can never be negative, and a shorter side can never exceed the hypotenuse - if either happens, the subtraction step needs checking.

# 5. The Converse and Pythagorean Triples

The theorem also works in reverse. That reverse statement is called its **converse**, and it is the tool for testing whether a triangle is right-angled when no angle is given.

**Converse of Pythagoras theorem:** If the square of the longest side of a triangle is equal to the sum of the squares of the other two sides, then the triangle is right-angled, and the right angle is opposite the longest side.

In symbols: if $a^2 + b^2 = c^2$ (c longest), the angle opposite side c is 90 degrees.

**Example 7.** Is a triangle with sides 9 cm, 12 cm and 15 cm right-angled?

Longest side is 15. Check: $9^2 + 12^2 = 81 + 144 = 225$, and $15^2 = 225$. The two are equal, so by the converse the triangle **is right-angled**, with the right angle opposite the 15 cm side.

**Example 8.** Is a triangle with sides 4 cm, 5 cm and 6 cm right-angled?

Longest side is 6. Check: $4^2 + 5^2 = 16 + 25 = 41$, but $6^2 = 36$. Since 41 is not equal to 36, the triangle is **not right-angled**. One failed equality settles the question - no angle needs to be measured.

## Pythagorean triples

Three whole numbers that satisfy $a^2 + b^2 = c^2$ are called a **Pythagorean triple**. Recognising them saves working time, because the third side can be written down at once.

| Triple (a, b, c) | Check | Useful multiples |
|---|---|---|
| 3, 4, 5 | 9 + 16 = 25 | 6, 8, 10 \| 9, 12, 15 \| 12, 16, 20 |
| 5, 12, 13 | 25 + 144 = 169 | 10, 24, 26 |
| 8, 15, 17 | 64 + 225 = 289 | 16, 30, 34 |
| 7, 24, 25 | 49 + 576 = 625 | 14, 48, 50 |
| 20, 21, 29 | 400 + 441 = 841 | - |

Any multiple of a triple is also a triple: doubling 3, 4, 5 gives 6, 8, 10, and 36 + 64 = 100 confirms it. In a question, divide the given sides by a common factor first - if they reduce to a known triple, the missing side follows immediately.

# 6. Applications - Finding the Hidden Triangle

Word problems in this unit all hide the same right-angled triangle. The skill is drawing it: a wall and the ground meet at 90 degrees, the sides of a rectangle meet at 90 degrees, and vertical and horizontal distances meet at 90 degrees. Once the triangle is drawn and the hypotenuse is named, the working is Sections 3 and 4 again.

**Example 9 - The ladder.** A ladder 13 m long leans against a wall. Its foot is 5 m from the wall. How high up the wall does it reach?

The ladder is the hypotenuse (13), the ground distance is one shorter side (5), and the height is the missing shorter side. Height$^2$ = 13$^2$ - 5$^2$ = 169 - 25 = 144, so the height = sqrt(144) = **12 m**.

**Example 10 - The rectangle diagonal.** A rectangular field is 9 m long and 12 m wide. Find the length of its diagonal.

The diagonal splits the rectangle into two right-angled triangles and is the hypotenuse. Diagonal$^2$ = 9$^2$ + 12$^2$ = 81 + 144 = 225, so the diagonal = sqrt(225) = **15 m**. The same method gives the diagonal of any rectangle or square: for a square of side s, the diagonal is s sqrt(2).

**Example 11 - Distance between two points on the ground.** A boy walks 6 km east and then 8 km north. How far is he from his starting point?

East and north are perpendicular, so the two walks are the shorter sides and the direct distance is the hypotenuse. Distance$^2$ = 6$^2$ + 8$^2$ = 36 + 64 = 100, so the distance = sqrt(100) = **10 km**. This is exactly the distance idea behind the distance formula in Unit 9.

**Example 12 - The broken tree.** A tree breaks 3 m above the ground. The top of the tree touches the ground 4 m from the foot of the tree. How long is the broken part?

The standing part (3 m) and the ground distance (4 m) are the shorter sides; the fallen, broken part is the hypotenuse. Broken part$^2$ = 3$^2$ + 4$^2$ = 9 + 16 = 25, so the broken part = sqrt(25) = **5 m**. The full height of the tree before it broke was 3 + 5 = 8 m.

**Three steps for every word problem:** (1) Sketch the situation and mark the right angle. (2) Label the hypotenuse - the side opposite that angle. (3) Add the squares for a missing hypotenuse, subtract for a missing shorter side, and answer in the units of the question.

# 7. Worked Mixed Examples

**Example 13.** In triangle ABC, angle B = 90 degrees, AB = 20 cm and BC = 21 cm. Find AC, the perimeter of the triangle, and its area.

AC is the hypotenuse: $AC^2 = 20^2 + 21^2 = 400 + 441 = 841$, so AC = sqrt(841) = **29 cm**. Perimeter = 20 + 21 + 29 = **70 cm**. Area = (1/2) x base x perpendicular = (1/2) x 21 x 20 = **210 $cm^2$**. For area, always use the two sides around the right angle - never the hypotenuse.

**Example 14.** Find the value of x: a right-angled triangle has hypotenuse (x + 2) cm and shorter sides x cm and 6 cm, and its hypotenuse is 10 cm.

Since the hypotenuse is 10, x + 2 = 10 gives x = 8. Check with the theorem: $x^2 + 6^2 = 64 + 36 = 100 = 10^2$. Correct, so **x = 8** and the sides are 6 cm, 8 cm and 10 cm.

**Example 15.** A rhombus has diagonals 10 cm and 24 cm. Find the length of one side.

The diagonals of a rhombus bisect each other at right angles, so each side is the hypotenuse of a right-angled triangle with shorter sides 5 cm and 12 cm (the halves of the diagonals). $Side^2 = 5^2 + 12^2 = 25 + 144 = 169$, so each side = sqrt(169) = **13 cm**.

**Example 16.** Prove that a triangle with sides 7 cm, 24 cm and 25 cm is right-angled, and state where the right angle lies.

Longest side is 25. $7^2 + 24^2 = 49 + 576 = 625 = 25^2$. By the converse of Pythagoras theorem, the triangle **is right-angled**, and the right angle is **opposite the 25 cm side**, between the sides of 7 cm and 24 cm.

## A routine for every question in this unit

1. Draw or picture the triangle and mark the 90-degree angle.
2. Name the hypotenuse - the side opposite that angle.
3. Write the formula first: $c^2 = a^2 + b^2$, rearranged if a shorter side is missing.
4. Substitute, square each length separately, then add or subtract.
5. Take the square root, keep the unit, and check the size: hypotenuse longest of all.

# 8. Formula Summary and Common Mistakes

| Situation | Rule |
|---|---|
| Pythagoras theorem | $c^2 = a^2 + b^2$, where c is the hypotenuse. |
| Find the hypotenuse | c = sqrt( $a^2 + b^2$ ) - add the squares. |
| Find a shorter side | a = sqrt( $c^2 - b^2$ ) - subtract the squares. |
| Test for a right angle (converse) | If $a^2 + b^2 = c^2$ with c longest, the triangle is right-angled opposite c. |
| Diagonal of a rectangle l by w | d = sqrt( $l^2 + w^2$ ). For a square of side s: d = s sqrt(2). |
| Known triples | 3-4-5, 5-12-13, 8-15-17, 7-24-25, 20-21-29, and all their multiples. |
| Area of the right-angled triangle | (1/2) x base x perpendicular - the two sides around the right angle. |

## Six mistakes that lose marks

1. Using the theorem on a triangle with no right angle. The theorem holds only for right-angled triangles.
2. Naming the wrong side as the hypotenuse. Always locate the 90-degree angle first and take the side opposite it.
3. Adding the squares when a shorter side is missing. A missing shorter side needs subtraction, and its answer must be smaller than the hypotenuse.
4. Forgetting the final square root: $c^2 = 100$ means c = 10, not c = 100.
5. Squaring wrongly, for example writing $6^2 = 12$ instead of 36, or working out $(a + b)^2$ as $a^2 + b^2$.
6. Taking roots too early: sqrt($a^2 + b^2$) is not a + b. With a = 6 and b = 8 the hypotenuse is 10, not 14.

> **One-line revision:** Hypotenuse squared equals the sum of the other two squares - add to find the hypotenuse, subtract to find a shorter side, and check equality to prove a right angle.

# 9. Practice - MCQs and Questions

## Multiple-choice questions

1. In a right-angled triangle, the side opposite the right angle is called the: (a) base (b) perpendicular (c) hypotenuse (d) diagonal
2. If the shorter sides are 3 cm and 4 cm, the hypotenuse is: (a) 5 cm (b) 6 cm (c) 7 cm (d) 25 cm
3. If the hypotenuse is 13 cm and one side is 5 cm, the other side is: (a) 8 cm (b) 12 cm (c) 18 cm (d) 144 cm
4. Which set of lengths can form a right-angled triangle? (a) 2, 3, 4 (b) 6, 8, 10 (c) 4, 5, 6 (d) 5, 6, 7
5. The diagonal of a square of side 1 unit is: (a) 1 (b) 2 (c) sqrt(2) (d) sqrt(3)
6. Pythagoras theorem can be applied only to: (a) any triangle (b) an equilateral triangle (c) a right-angled triangle (d) an isosceles triangle

## Short and long questions

1. Find the hypotenuse of a right-angled triangle whose shorter sides are 9 cm and 12 cm.
2. The hypotenuse of a right-angled triangle is 26 cm and one shorter side is 10 cm. Find the other shorter side.
3. Check whether triangles with these sides are right-angled: (a) 8 cm, 15 cm, 17 cm (b) 5 cm, 7 cm, 9 cm.
4. A ladder 15 m long reaches a window 12 m above the ground. How far is the foot of the ladder from the wall?
5. A rectangular park is 30 m by 40 m. Find the length of the diagonal path across it, and the perimeter of the park.

## Answer key

**MCQs:** 1(c), 2(a), 3(b), 4(b), 5(c), 6(c).

**Questions:** 1) sqrt(81 + 144) = sqrt(225) = 15 cm. 2) sqrt(676 - 100) = sqrt(576) = 24 cm. 3) (a) 64 + 225 = 289 = $17^2$, so right-angled. (b) 25 + 49 = 74, not 81 = $9^2$, so not right-angled. 4) sqrt(225 - 144) = sqrt(81) = 9 m from the wall. 5) Diagonal = sqrt(900 + 1600) = sqrt(2500) = 50 m; perimeter = 2 x (30 + 40) = 140 m.

> **Final tip:** In the board paper, draw the triangle, mark the right angle, and write the formula before any numbers. In this unit the add-or-subtract decision is worth as much as the arithmetic that follows it.